
ASSIGNMENT - 3

COMPUTER NETWORKS (PCCST501)

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Class: S5 CSE A

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LAN ,ETHERNET & WIRELESS NETWORK

INVESTIGATION

Question 1 – Explain the responsibilities of the data-link layer and distinguish logical link control from media access control.

Answer:

Data-Link Layer Responsibilities

The primary function of the Data-Link Layer is node-to-node (hop-by-hop) delivery of data across a single physical link connecting adjacent nodes. Its main responsibilities include:

1. **Framing:** Grouping the raw stream of bits received from the physical layer into distinguishable data units called frames.
2. **Flow Control:** Regulating the rate and amount of data transmission to prevent a fast sender from overwhelming a slow receiver.
3. **Error Control:** Detecting and recovering from corrupted, lost, or duplicated frames via mechanisms like checksums, CRCs, and retransmission protocols.
4. **Addressing (Node-to-Node):** Adding physical (MAC) source and destination addresses to each frame header.

Feature	Logical Link Control (LLC)	Media Access Control (MAC)
Position	Upper sublayer of the Data Link Layer.	Lower sublayer of the Data Link Layer.
Primary Role	Provides link control functions common to all media types.	Handles media access control specific to shared or broadcast links.
Key Functions	Flow control, error control, framing part, and protocol multiplexing.	Framing, MAC addressing, and channel access control (CSMA/CD, CSMA/CA).
Hardware/Media Independence	Independent of physical topology and medium.	Specific to the underlying physical medium and network topology.

Table 1.1: Logical Link Control (LLC) vs Media Access Control (MAC)

Question 2 – Explain Ethernet frame format and identify the purpose of destination MAC, source MAC, type/length and FCS fields.

Answer:

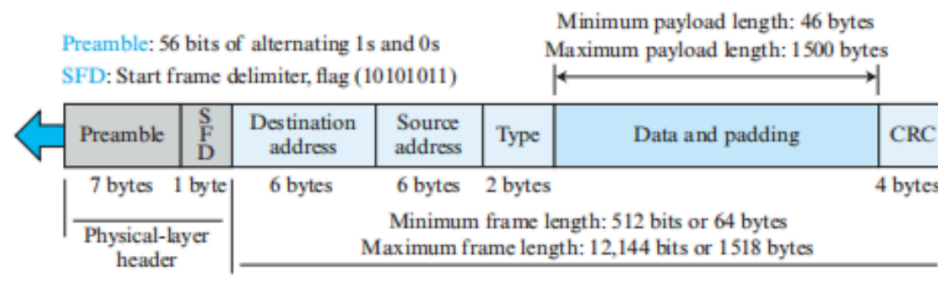


Fig 2.1: Ethernet Frame Format

- **Destination MAC Address (6 Bytes / 48 bits):** Specifies the physical address of the intended receiving station or network interface card (NIC).
- **Source MAC Address (6 Bytes / 48 bits):** Specifies the physical address of the station generating and transmitting the frame.
- **Type/Length (2 Bytes):** In Ethernet II, this field specifies the network-layer protocol encapsulated in the frame body (e.g., 0x0800 for IPv4, 0x0806 for ARP). In IEEE 802.3, values ≤ 1500 indicate the frame payload length in bytes.
- **Frame Check Sequence (FCS) (4 Bytes):** Contains a 32-bit Cyclic Redundancy Check (CRC-32) value calculated over the frame header and payload. It enables the receiver to detect bit errors caused by transmission noise and discard corrupted frames.

Question 3 – Use Wireshark to capture Ethernet traffic and identify source and destination MAC addresses from at least five frames.

Answer:

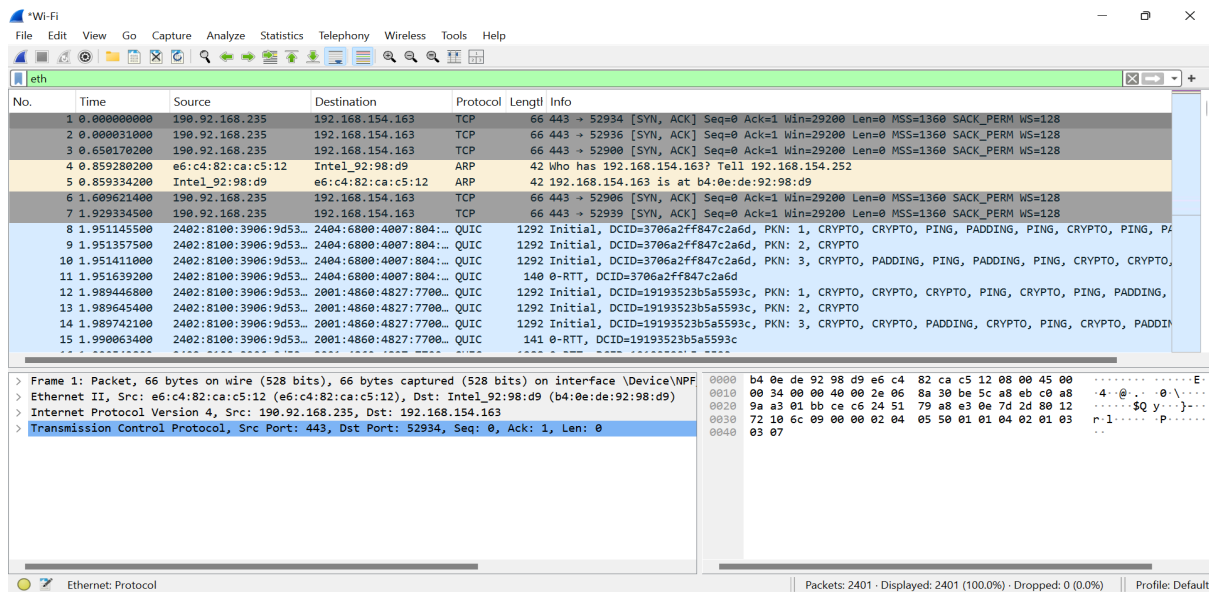


Fig 3.1: Wireshark Ethernet Packet Capture

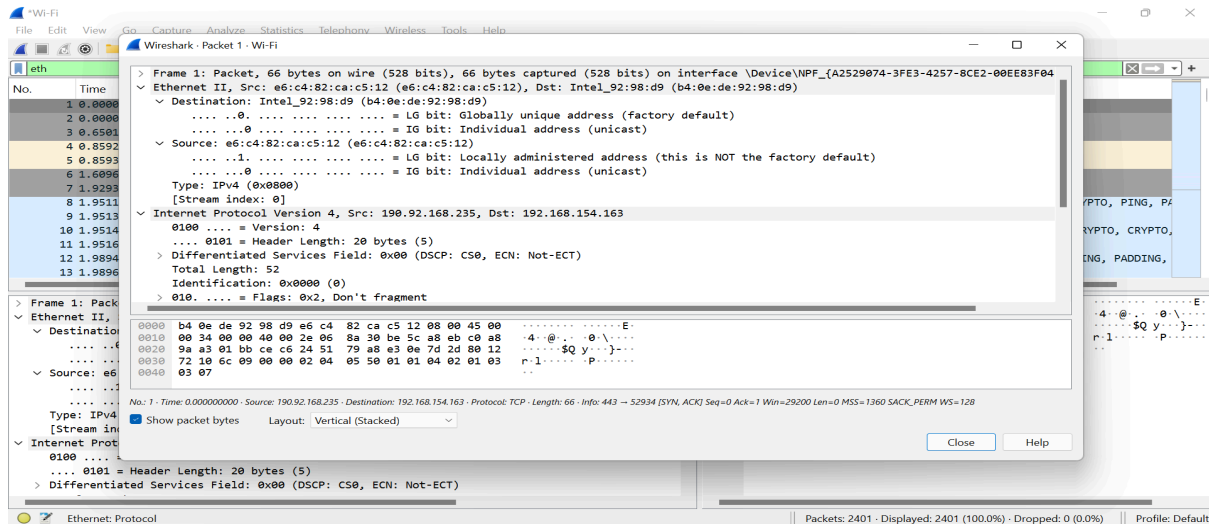


Fig 3.2: Packet 1

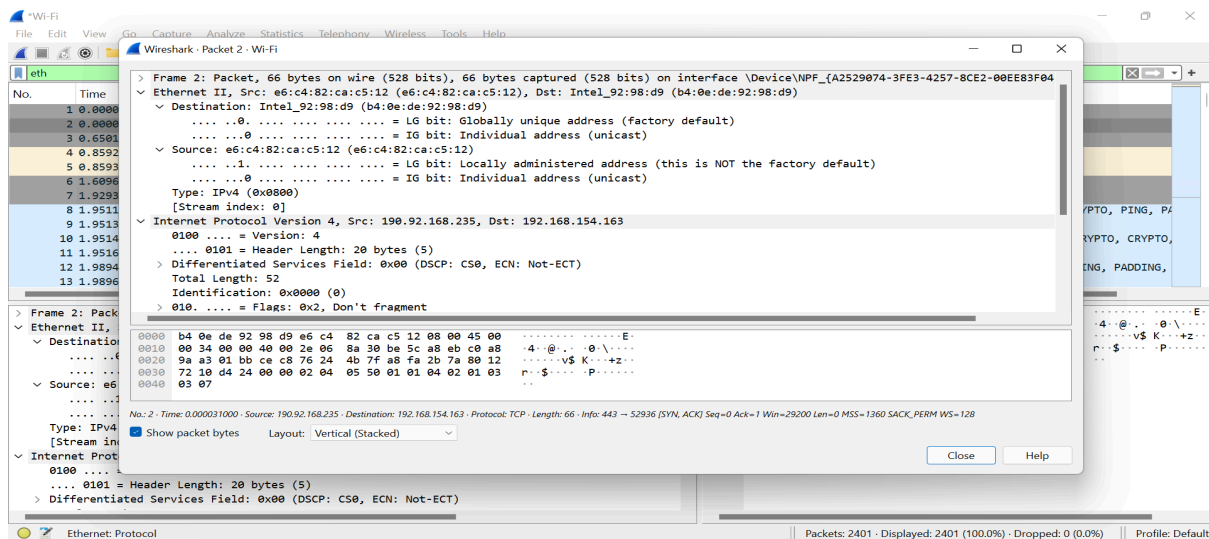


Fig 3.3: Packet 2

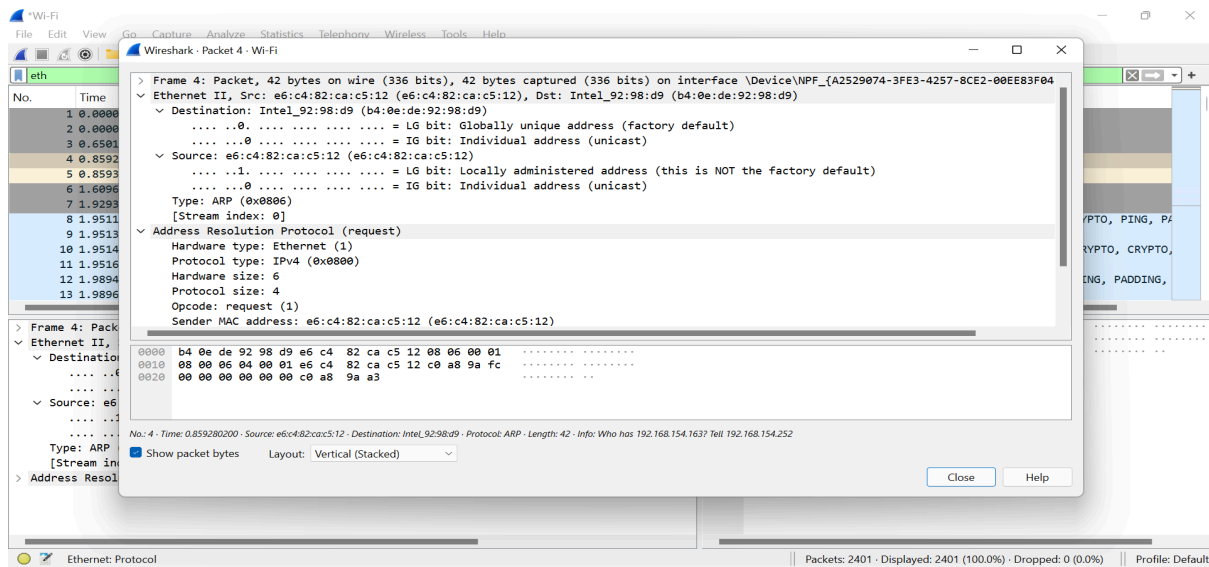


Fig 3.4: Packet 3

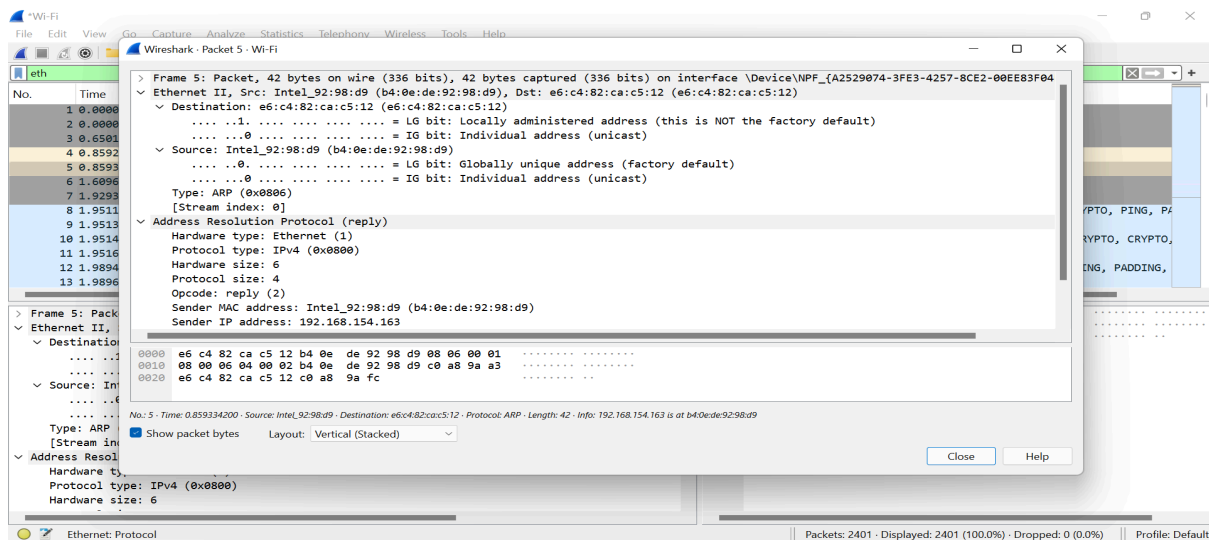


Fig 3.5: Packet 4

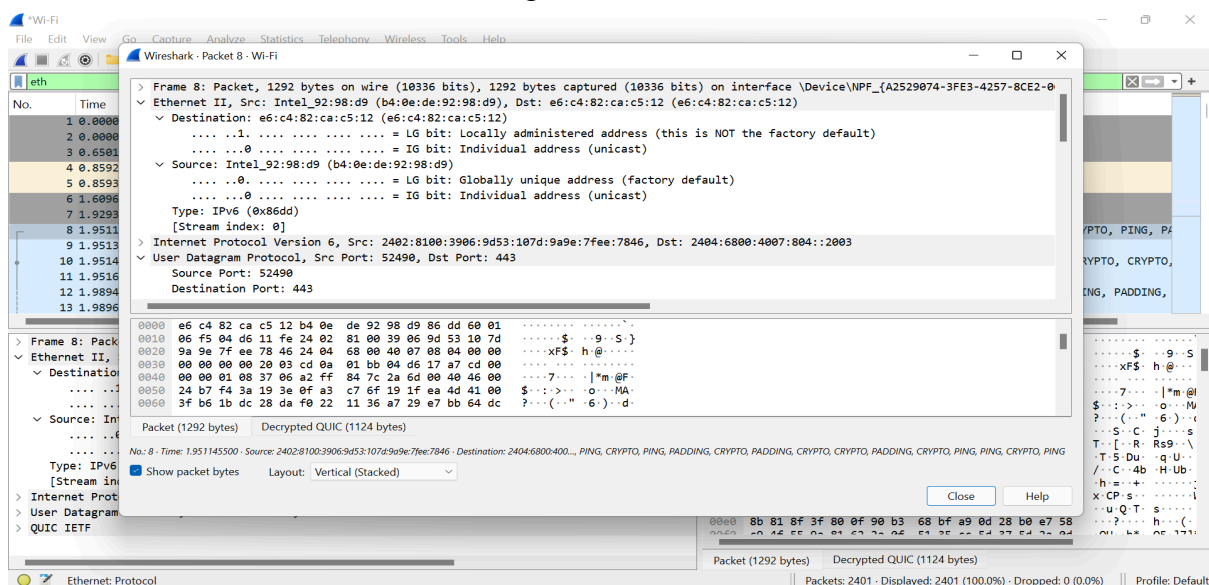


Fig 3.6: Packet 5

Frame No.	Packet No.	Source MAC Address	Destination MAC Address	Protocol
1	1	e6:c4:82:ca:c5:12	b4:0e:de:92:98:d9	IPv6
2	2	e6:c4:82:ca:c5:12	b4:0e:de:92:98:d9	IPv4
3	4	e6:c4:82:ca:c5:12	b4:0e:de:92:98:d9	ARP
4	5	b4:0e:de:92:98:d9	e6:c4:82:ca:c5:12	ARP
5	8	b4:0e:de:92:98:d9	e6:c4:82:ca:c5:12	IPv6

Table 3.1: Ethernet Frame Analysis

Question 4 – Explain the operation of a switch using MAC address learning. Demonstrate how a switch builds its forwarding table.

Answer:

Switch Operation Principle

A link-layer switch forwards frames based on physical (MAC) addresses using a dynamic lookup table called a **Switching Table** or **MAC Address Table**.

1. **Learning:** When a frame arrives at an ingress port, the switch inspects its **Source MAC Address**. It maps that address to the incoming port number and records/refreshes the entry in its switching table.
2. **Forwarding / Filtering / Flooding:** The switch inspects the frame's **Destination MAC Address**:
 - **Unicast Known:** If the destination MAC is present in the table, the frame is forwarded only through the specific mapped port.
 - **Unicast Unknown:** If the destination MAC is not in the table, the switch floods the frame out of all active ports except the incoming port.

- **Broadcast/Multicast:** Frames addressed to broadcast (FF:FF:FF:FF:FF:FF) or multicast addresses are automatically flooded to all ports except the source port.

MAC Forwarding Table Construction

Step 1

- Station A sends frame to Station D
- Switch does not know A or D
- Frame is flooded
- Learns: A is connected to Port 1

Step 2

- Station D sends frame
- Switch learns D's port

Result

- Table gradually becomes complete

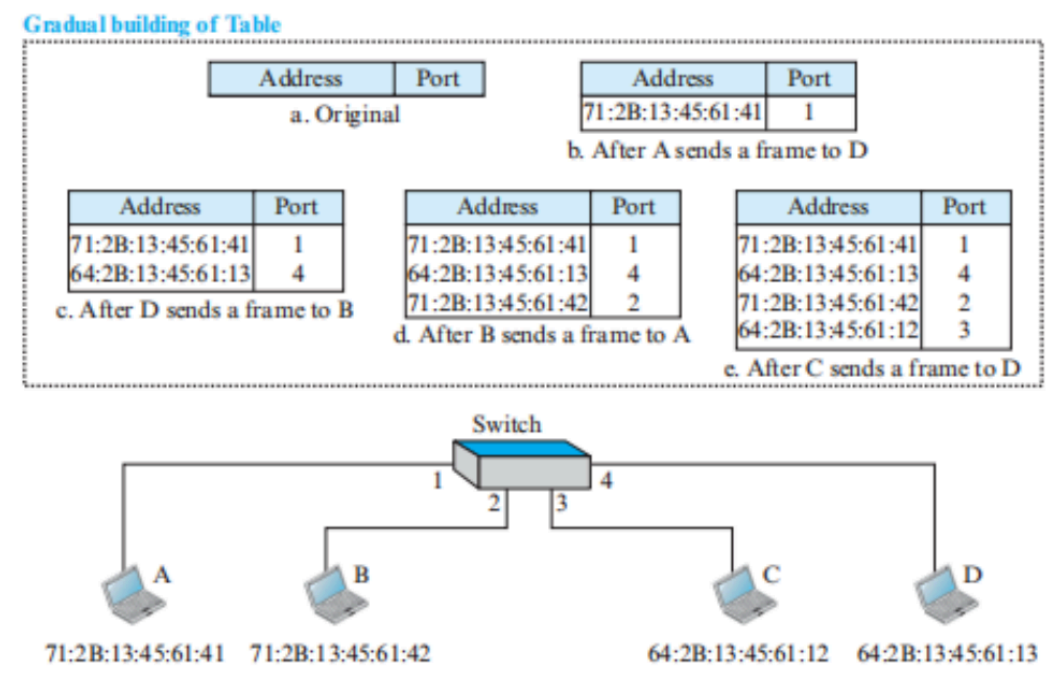


Fig 4.1: Learning Switch

Question 5 – Compare hub, switch and bridge in terms of forwarding, collision domains and practical use.

Answer:

Characteristic	Hub	Bridge	Switch
Layer of Operation	Physical Layer (Layer 1)	Data-Link Layer (Layer 2)	Data-Link Layer (Layer 2)
Forwarding Method	Broadcasts incoming bits to all other ports	Selective forwarding based on MAC addresses	Selective forwarding using a hardware MAC address table
Collision Domains	One shared collision domain across all ports	Separates collision domains per segment	Separates collision domains per port (micro-segmentation)
Broadcast Domains	One single broadcast domain	One single broadcast domain	One single broadcast domain per VLAN
Duplex Mode	Half-duplex only	Half-duplex or full-duplex	Full-duplex dedicated links
Practical Use	Legacy or obsolete star topologies	Connecting a few LAN segments	Modern Ethernet LAN aggregation and enterprise networks

Table 5.1: Hub vs Bridge vs Switch

Question 6 – Explain CSMA/CD and discuss why its importance is reduced in modern switched full-duplex Ethernet.

Answer:

CSMA/CD Protocol Operation

CSMA/CD (Carrier Sense Multiple Access with Collision Detection) manages shared media access:

1. **Carrier Sense (Listen Before Talk):** A station checks if the physical channel is idle before transmitting.
2. **1-Persistent Strategy:** If busy, it waits until idle; when idle, it transmits immediately.
3. **Collision Detection (Listen While Talk):** While transmitting, the station monitors energy levels on the cable. A sudden increase in energy indicates a collision.
4. **Abortion and Jamming:** If a collision occurs, the station stops frame transmission, sends a 48-bit **jam signal** to notify all hosts, increments retry count K , and calculates a backoff time using Binary Exponential Backoff.
5. **Backoff Formula:** $T_B = R \times T_{fr}$, where R is a random integer in range $[0, 2^K - 1]$ (for $K \leq 10$, $\max K = 15$).

Why CSMA/CD Importance is Reduced in Modern Networks

In modern Fast/Gigabit/10G Switched Ethernet:

- **Micro-Segmentation & Private Media:** Each host is connected via a dedicated point-to-point link directly to a switch port.
- **Full-Duplex Communication:** Independent transmit (Tx) and receive (Rx) wire pairs/fiber channels allow simultaneous bidirectional transmission.
- **Zero Collisions:** Because signals travel over physically separate channels without shared contention, collisions are mathematically impossible, completely eliminating the need for CSMA/CD.

Question 7 – Explain CSMA/CA in IEEE 802.11 and compare its objective with CSMA/CD.

Answer:

CSMA/CA Operation in Wireless LANs

Wireless nodes use CSMA/CA (Collision Avoidance) because wireless transceivers cannot perform collision detection (signals degrade over distance, and node transceivers cannot transmit and listen at the same time):

1. **Carrier Sensing:** A node senses both physical medium energy and virtual channel reservation status via the **Network Allocation Vector (NAV)**.
2. **IFS (Interframe Space):** Once the channel becomes free, the node waits for an Interframe Space (e.g., **DIFS** for standard data).
3. **Contention Window:** The node selects a random backoff time slot within a window $[0, 2^K - 1]$ to decrement while the channel remains idle.
4. **RTS/CTS Handshake (Optional):** To eliminate the **Hidden Station Problem**, the sender transmits a **Request to Send (RTS)** frame. The receiver responds with a **Clear to Send (CTS)** frame. Nearby stations reading RTS/CTS adjust their NAV timers to defer transmission.
5. **Explicit Acknowledgements (ACK):** After receiving data, the destination waits **SIFS** and returns an explicit **ACK** frame. Wireless transmissions are considered failed if an ACK is not received.

Parameter	CSMA/CD (Ethernet)	CSMA/CA (IEEE 802.11)
Primary Goal	Collision Detection and recovery	Collision Avoidance before transmission
Transmission Monitoring	Listens continuously during bit transmission	Cannot reliably listen for collisions while transmitting
Core Mechanism	Stop transmission, send a jamming signal, and use exponential backoff	IFS (DIFS/SIFS), contention window, RTS/CTS, and explicit ACK

Medium Type	Wired shared copper channels	Wireless shared RF spectrum
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Table 7.1: CSMA/CD vs CSMA/CA

Question 8 – Design a small wireless LAN for a college laboratory. Specify access point placement, SSID, channel considerations, addressing and basic security requirements.

Answer:

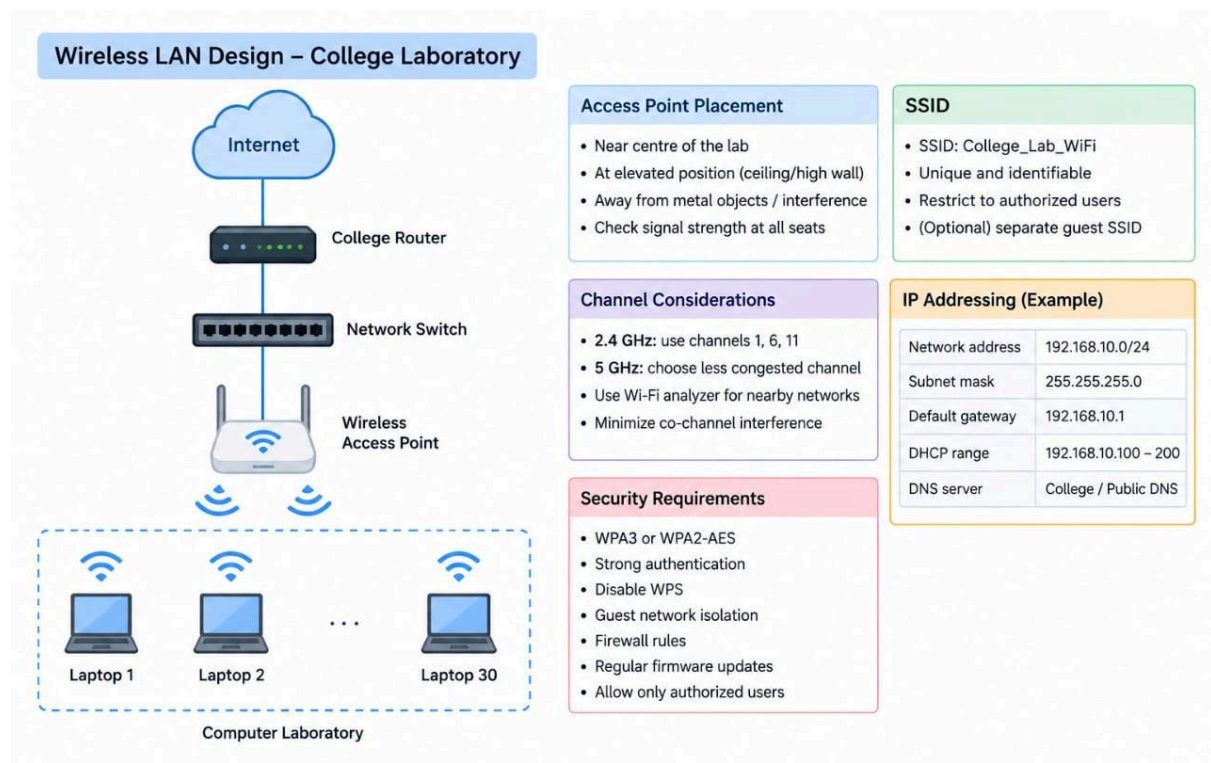


Fig 8.1: College Library Wireless LAN Design

Question 9 – Explain Mobile IP and the need for mobility support when a device moves between networks.

Answer:

Mobile IP

Core Components & Addresses

- **Mobile Host (MH):** Moving device that changes its network attachment point.
- **Home Address:** Permanent IP address assigned to the MH on its home network.
- **Care-of Address (COA):** Temporary IP address acquired by the MH on a foreign network.
- **Home Agent (HA):** Router on the home network that intercepts packets for the MH and tunnels them to the COA.
- **Foreign Agent (FA):** Router on the visited network that receives tunneled packets, decapsulates them, and delivers them to the MH.

Three Operational Phases

1. **Agent Discovery:** The MH discovers its network location via periodic ICMP Agent Advertisements or Agent Solicitations, acquiring a Care-of Address.
2. **Registration:** The MH sends a Registration Request to the FA, which relays it to the HA. The HA updates its binding table and sends a Registration Reply back.
3. **Data Transfer:**
 - **Incoming Traffic:** Packets sent to the MH's Home Address are intercepted by the HA via Proxy ARP, encapsulated (tunneled) to the COA, decapsulated by the FA, and delivered to the MH.
 - **Outgoing Traffic:** Packets from the MH to the Remote Host are sent directly using standard IP routing (Triangular Routing).

Need for Mobility Support

Standard IPv4 routing relies on fixed network prefixes. If a device moves to a new network keeping its original IP address, packets cannot be routed to it. Changing the IP address breaks active TCP socket connections. Mobile IP resolves this by separating identity from location.

Question 10 – Perform a packet-level comparison of a wired Ethernet frame and a wireless LAN frame using suitable captures or documented examples.

Answer:

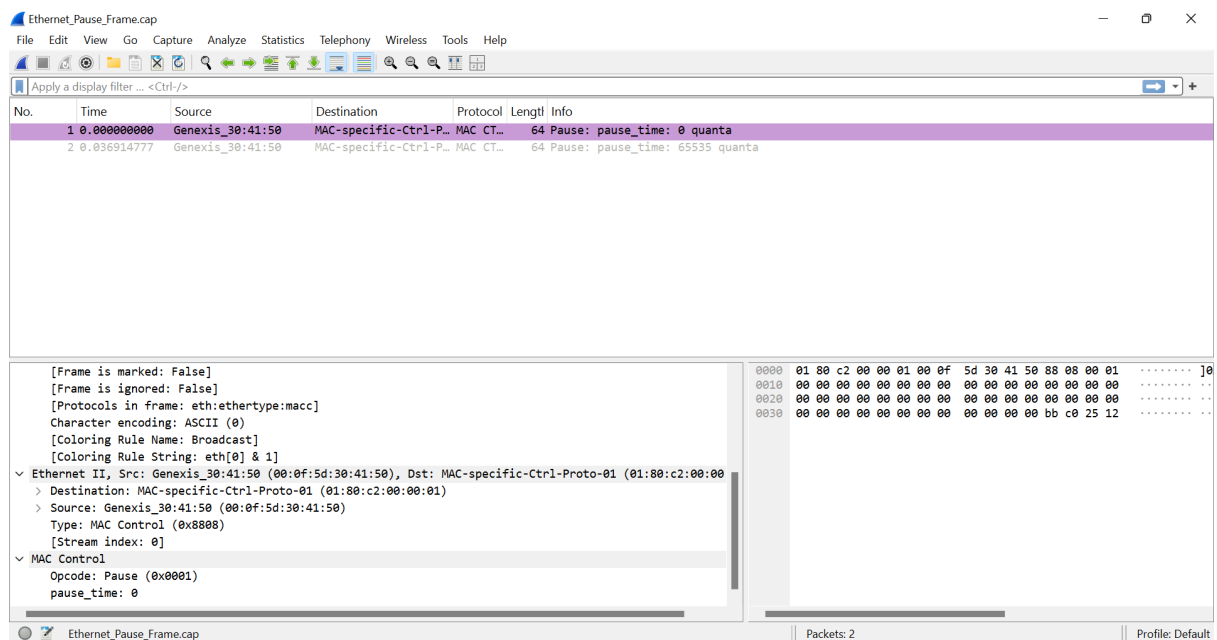


Fig 10.1: Ethernet Pause Frame

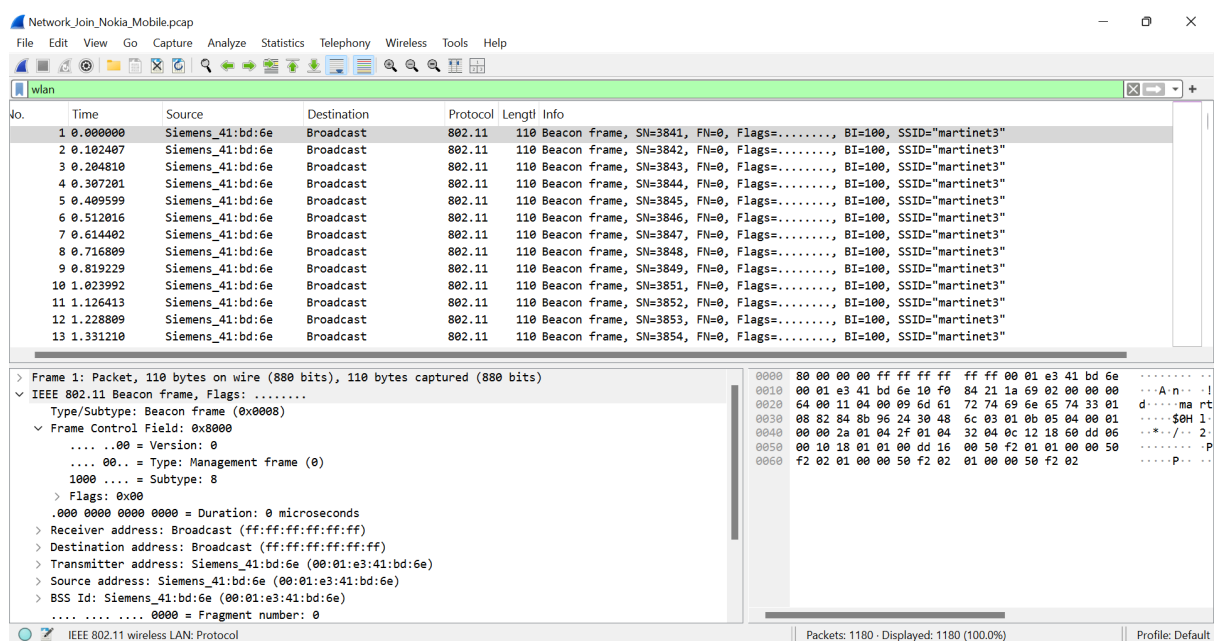


Fig 10.2: Wireless LAN Beacon Frame

Feature	Wired Ethernet Frame	Wireless LAN Frame
Technology	Ethernet	IEEE 802.11
Communication medium	Wired cable	Wireless radio waves
Captured frame type	MAC Control Pause frame	Beacon frame
Frame length	64 bytes	110 bytes
Source MAC address	00:0f:5d:30:41:50	00:01:e3:41:bd:6e
Destination address	01:80:c2:00:00:01	ff:ff:ff:ff:ff:ff (Broadcast)
Type / Frame Control	Type: MAC Control (0x8808)	Frame Control: 0x8000
Additional fields observed	Opcode: Pause (0x0001), Pause time: 0	BSS ID: 00:01:e3:41:bd:6e, SSID: martinet3, Duration: 0 microseconds
Purpose of captured frame	Used for Ethernet flow control	Announces the wireless network's presence and information
Addressing structure	Uses source and destination MAC addresses	Can contain multiple address fields, depending on the frame type

Table 10.1: Wired Ethernet Frame vs Wireless LAN Frame